

Spark Gap upgrade (version 0.3)

This document is a how to build a Spark Gap that takes pure Zinc rods.

Listing of the components needed from Amazon when you want to build this Spark Gap.

99.95% pure Zinc rods (Diameter 10mm):

https://www.amazon.nl/dp/B0C8692WW4?ref=ppx_yo2ov_dt_b_fed_asin_title

SK10 Aluminum Linear Motion Rail Clamp Guide, Support 10mm Diameter Shaft

Expensive better looking:

https://www.amazon.nl/dp/B0D6FWW1QW?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Cheaper version:

https://www.amazon.nl/dp/B0C477VJZY?ref=ppx_yo2ov_dt_b_fed_asin_title

External junction box, junction box, IP65 waterproof, project box, external electrical project boxes (black transparent, 20 x 12 x 7.5 cm)

https://www.amazon.nl/dp/B0CZ8WJDJP?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Or the gray version:

https://www.amazon.nl/dp/B0B2MJH7Q5?ref=ppx_yo2ov_dt_b_fed_asin_title

Copper Plate T2, 3mm x 15mm x 250mm Cu Metal Flat Bar:

https://www.amazon.nl/dp/B09ZXVJCWR?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

WINSINN 40mm Color LED Fan 12V Hydraulic Bearing Brushless 4010 40x10mm - High Speed (Pack of 4 pieces):

https://www.amazon.nl/dp/B087T5PC2F?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Fan Grill 40mm Fan Cover

https://www.amazon.nl/dp/B0D9GT7NSB?ref=ppx_yo2ov_dt_b_fed_asin_title

HEXATON Cylinder head screws Cylinder head bolts with hex M6 x 20 mm:

https://www.amazon.nl/dp/B09LHX76CX?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Hexagon nut - M6 - Galvanized steel - DIN 934 – C45712:

https://www.amazon.nl/dp/B08N1BXZP9?ref=ppx_yo2ov_dt_b_fed_asin_title

QUARKZMAN M3 Male to Female Hexagonal Standoffs x 50pcs Hexagonal Motherboard Nut Coupler for Computer and Circuit Board:

https://www.amazon.nl/dp/B0CG9QFKRY?ref=ppx_yo2ov_dt_b_fed_asin_title

PC Screws M3 x 1/4 Inch - Long Spacers - Pack of 50 - Computer Screws:

https://www.amazon.nl/-/en/gp/product/B00032Q1J4/ref=ewc_pr_img_1?smid=A17D2BRD4YMT0X&psc=1

There are other things I got at the local hardware store.

Plexiglass 4mm thick

Plate-screws 4.2 x 13mm

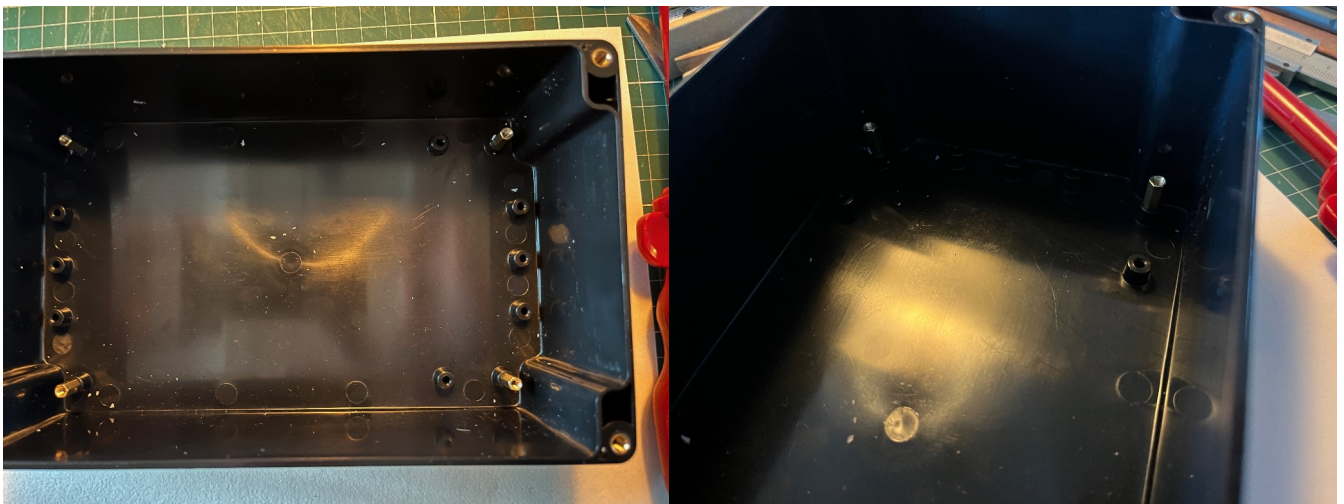
M6 washers.

Shrink sleeve (red and black) (optional)

Tools needed, Philips screw driver, tapping set, feeler gauge or boxcutter, saw (to cut the Plexiglass to size), drill, drill bit set, hot air gun (for the shrink sleeve), center punch, masking tape, hole saw that goes on the drill.

The plate that is going to hold the Spark Gap:

Cut a rectangular piece out of the 4 mm plexiglass which is 11 cm by 17 cm. This will fit perfect in the junction box and leaves room for the 2 fans on either side.

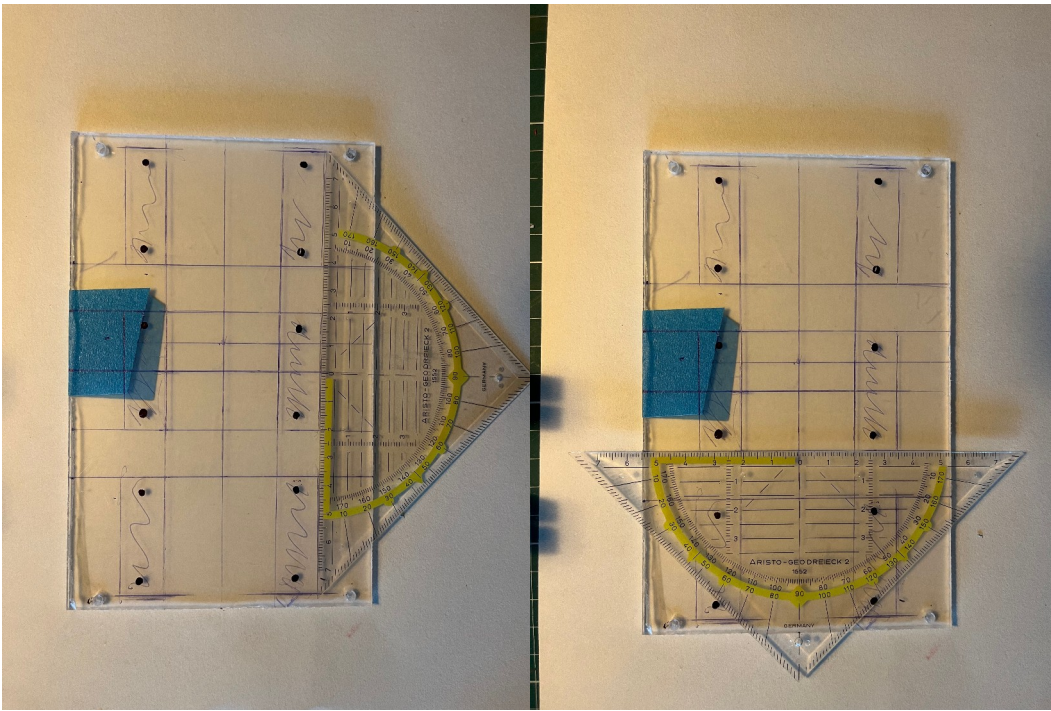


In the 4 corners of the Junction Box there are 4 nobs with a hole, tap the hole M3 so it will take the PCB riser studs.

Place the plexiglass in the box and mark the place where holes need to go for the PXC screws (Forgot to take a photo)

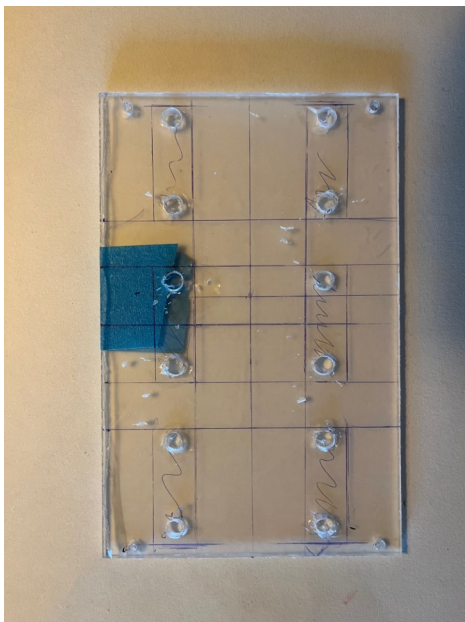
Draw out the places where the SK10 Aluminum Linear Motion Rail Clamp will go.

First do the ones in the center and from there do the right and left side so your Spark Gap will look neat and symmetrical. Distance between the center and the outer SK10 places is 16 mm.



Note: the measurements in the images above are in centimeters.

On the protective foil of plexiglass I have drawn the positions where my SK10 Rail Clams will go as well as where I have to drill the holes for the screws.

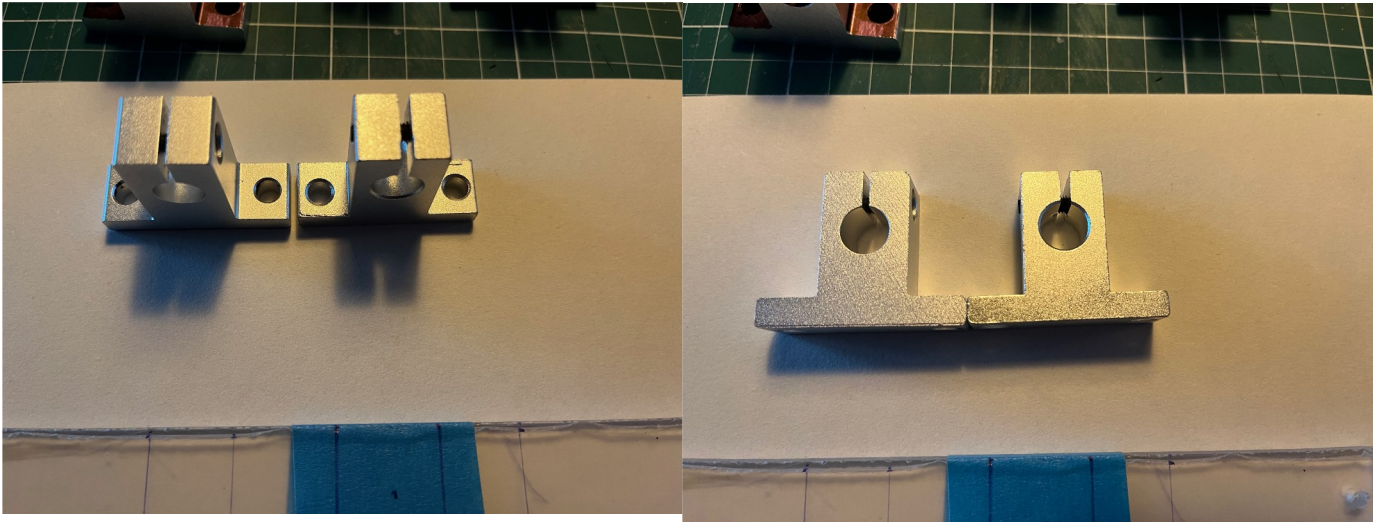


Drill holes for the bolts that hold the Zinc rod holders (drill needed 6.5 mm)

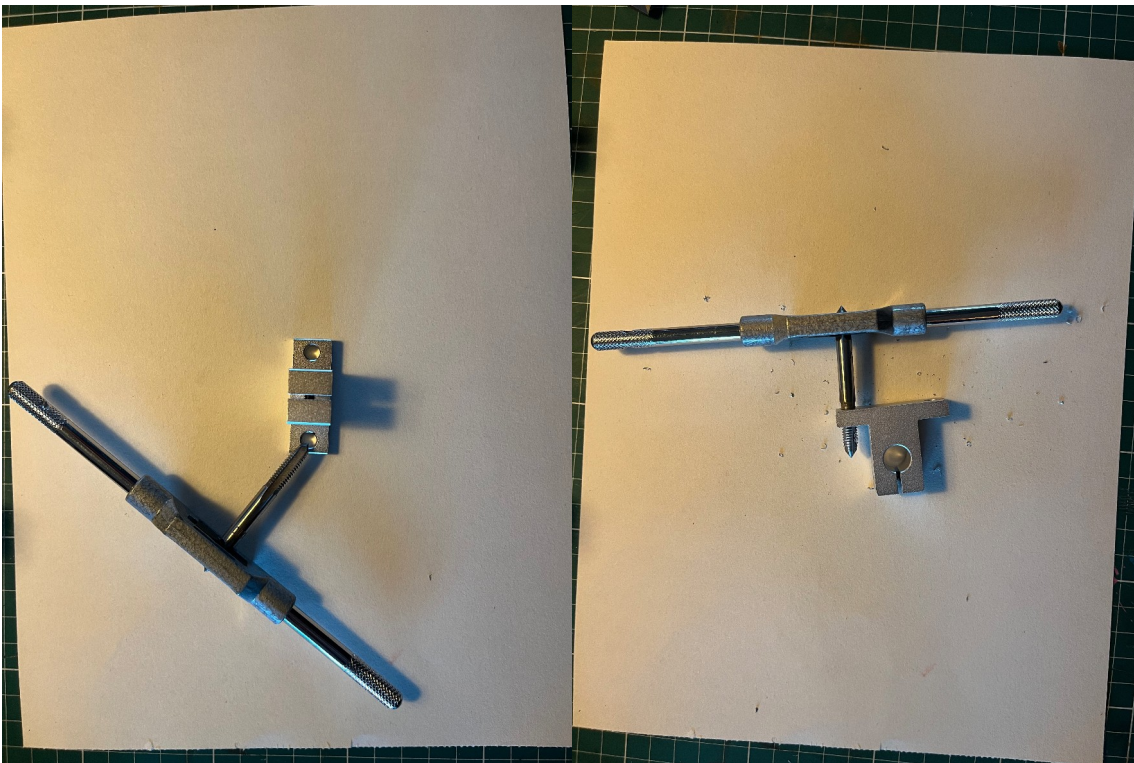
Drill the holes for the PC screws (drill needed 3.5 mm)

SK10 Rail Clamps and what needs to be done before they can be used as the holders for the pure zinc rods (diameter 10 mm)

There are several versions of the SK10 Rail clamps, some are more messy looking but both versions will work for the Spark Gap (see photos below)

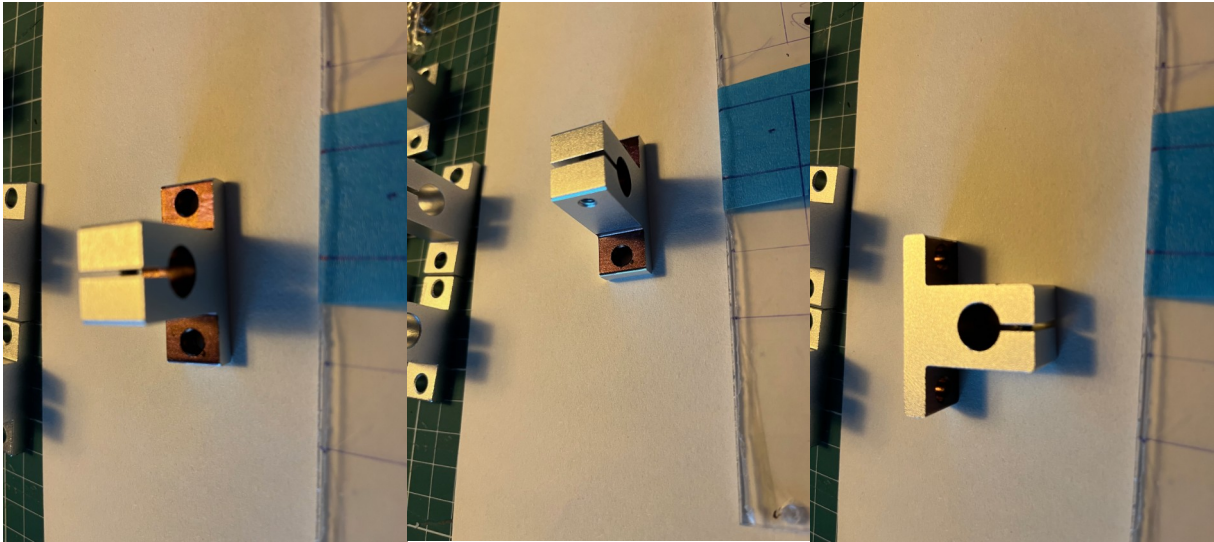


First step is: thread the holes in the SK10 (M6 tap) best way to put in the threading is from the bottom.



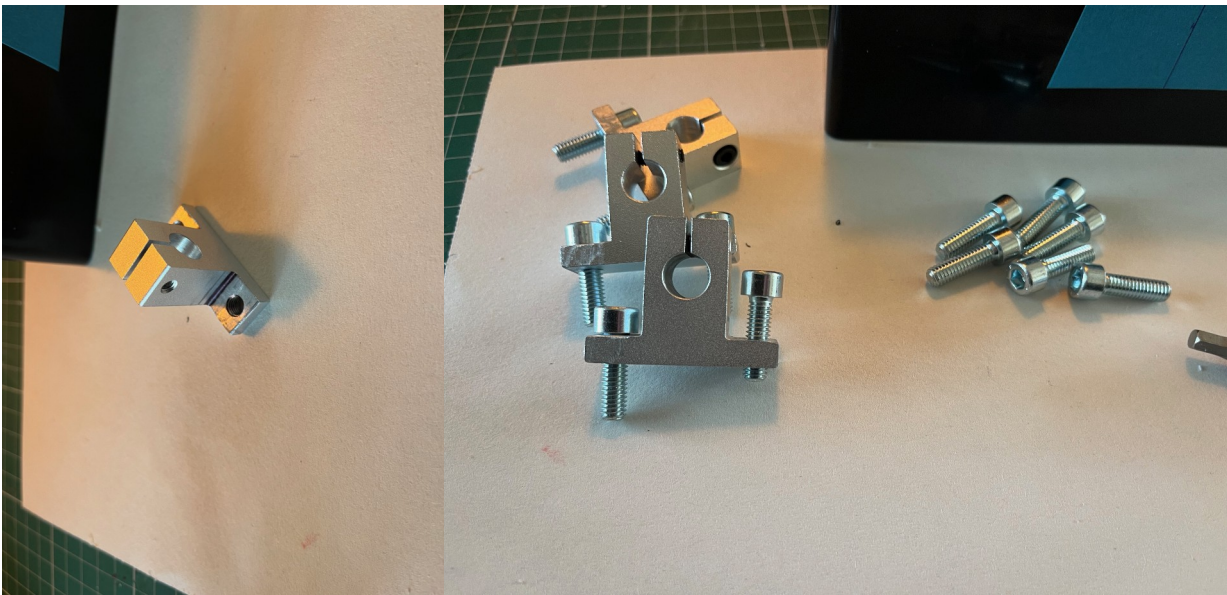
The holes will be tapped so they will take the M6 screws and that will also take away the protective coating.

Next is sanding down the spots that need to conduct electricity.

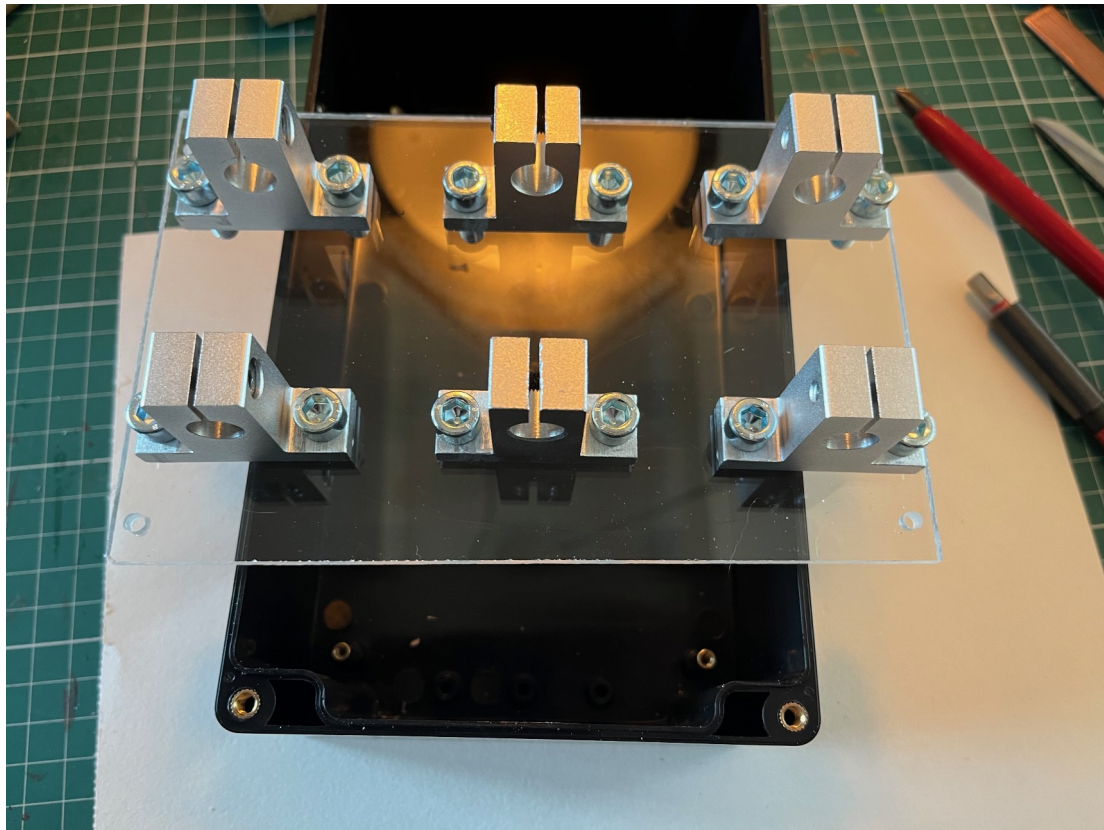


In the images above you can see on of the clamps, the parts that have been made black with a permanent marker has to be sanded down. As the aluminum has a protective layer that makes it so it doesn't conduct electricity.

NOTE: Do not drill or sand the holes you taped in the first part. Threading the holes already removed the protective coating.



After threading the holes and sanding down the protective layer it is time to screw in the M6 bolts. Now you can place the SK10's on the plexiglass, if the fit is to tight you can drill up the holes to 7mm in the plexiglass.



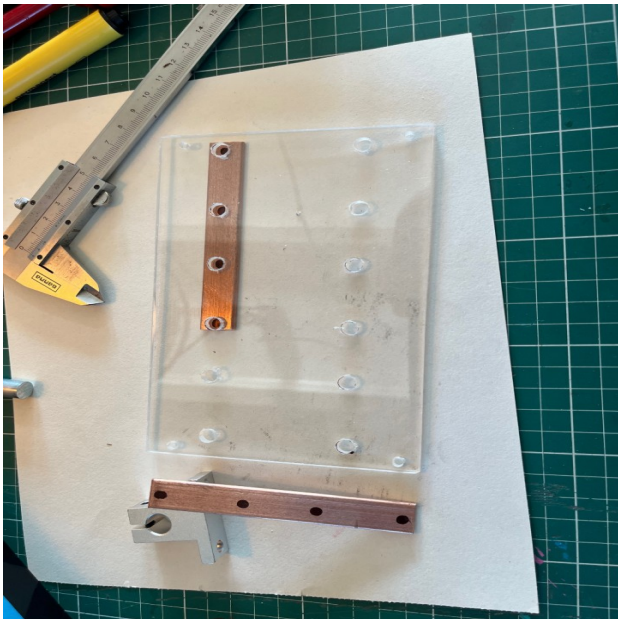
Cutting the Copper bar and the Zinc rods

First measure the length that is needed for the copper bar (if you used the 16mm distance between the center SK10 studs and the outer the length of the copper bar you need is 10cm).

The pure Zinc rods are 10cm and need to be cut in half mark the rods at 5cm.

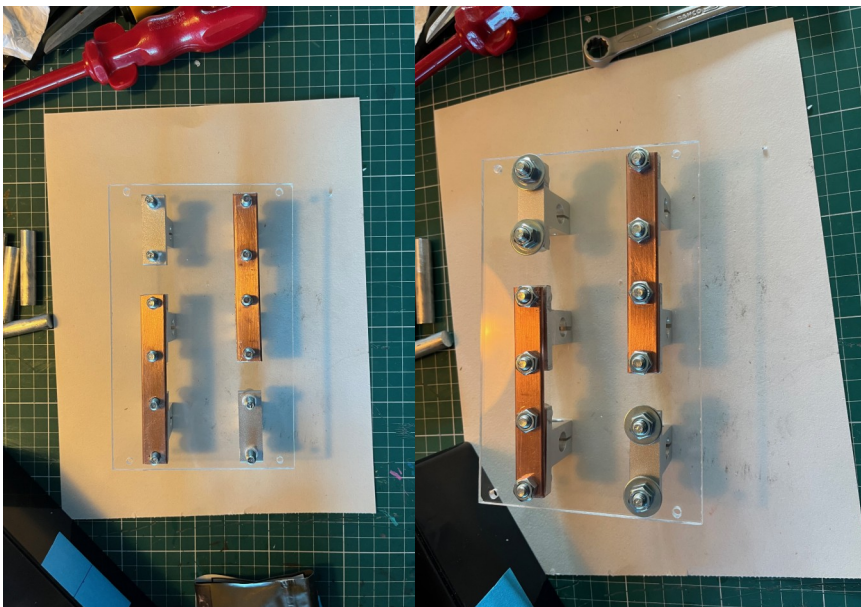


Note: Both Copper and Zinc are good heat conductors. If you cut them they can heat up really quickly.



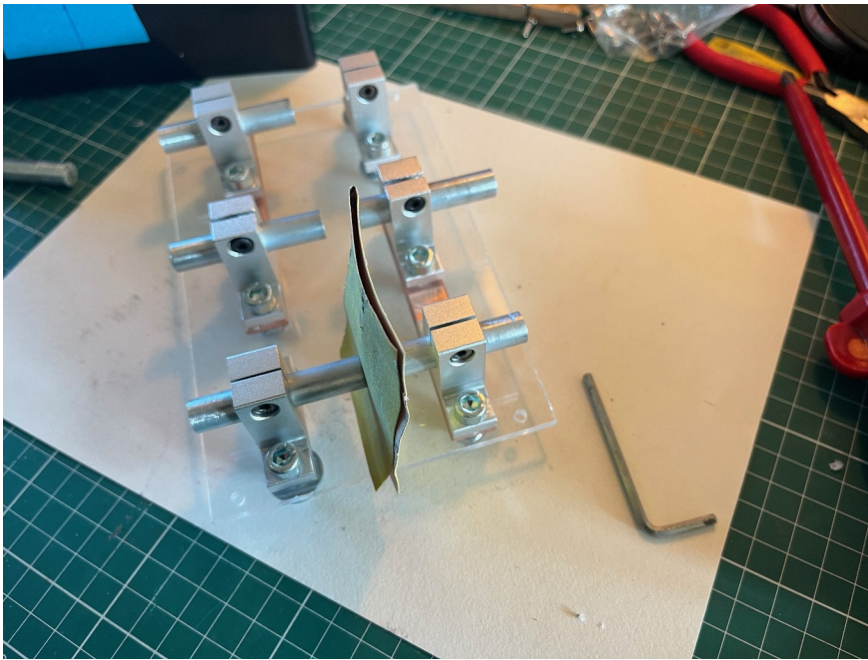
Now use the plexiglass to mark the Copper bars so you know where you need to drill holes. Mark the center of the bar and use the center punch to hit a dent on where the holes need to be drilled.

Drill the holes in the copper bars, use a 6.5 mm drill bit. Again be aware that the Copper will heat up a lot.



After drilling the holes you can see if the Copper fit correctly, if it is to tight you can drill the holes up to 7mm.

If everything fits perfectly you can put on the M6 nuts and tighten everything up. On the outer SK10's place washers before putting on the nuts.



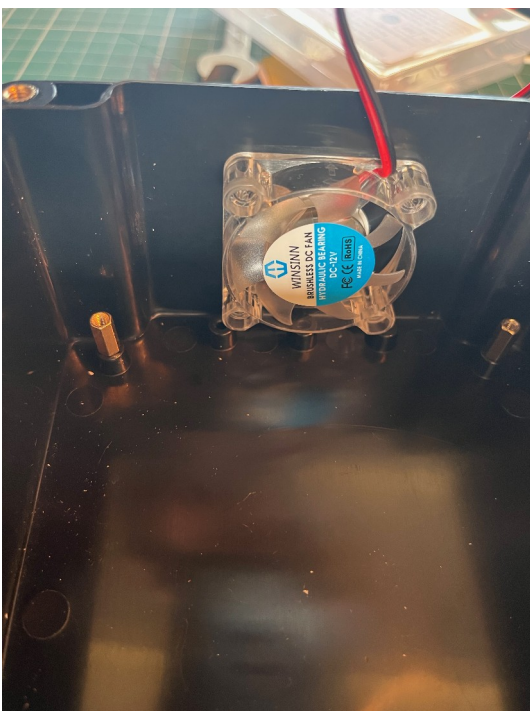
Now you can slide in the Zinc rods and set them at the correct distance.

First fixate one side and then use a feeler gauge or boxcutter (thickness of the blade) to the correct Spark Gap distance (0.4 to 0.5 mm) and fixate the other side.

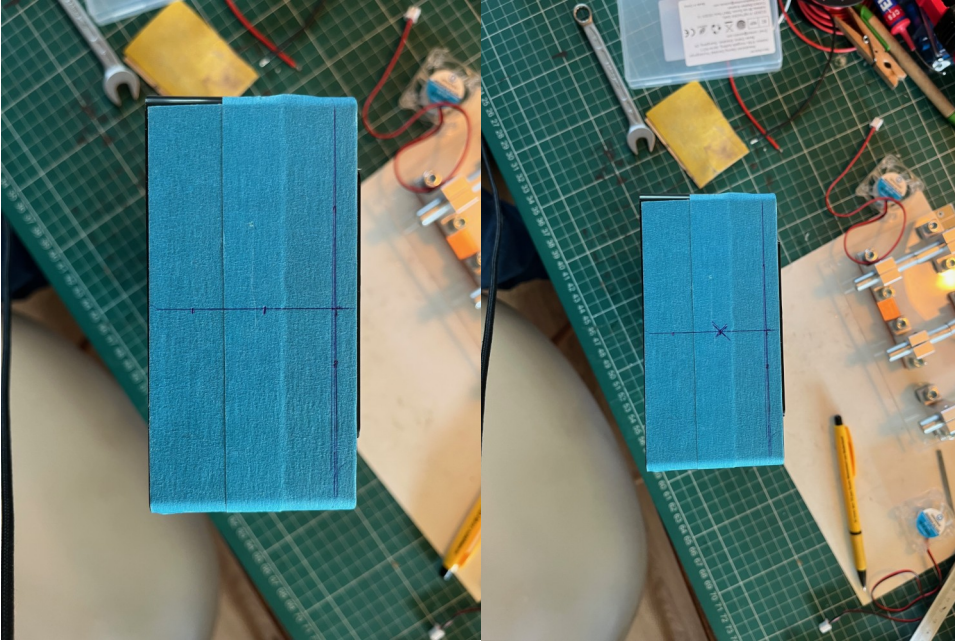
To smooth out the surface you can use a piece of sanding paper.

Installing the 40mm fans

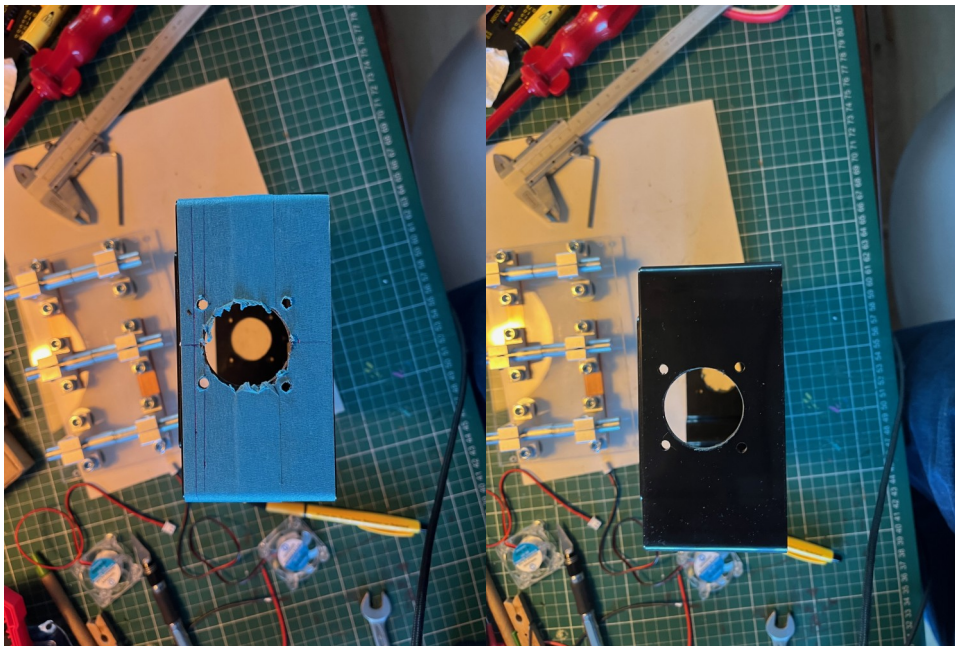
First you have to mark where to place the fans, they shouldn't be placed too low in the box (see photo below)



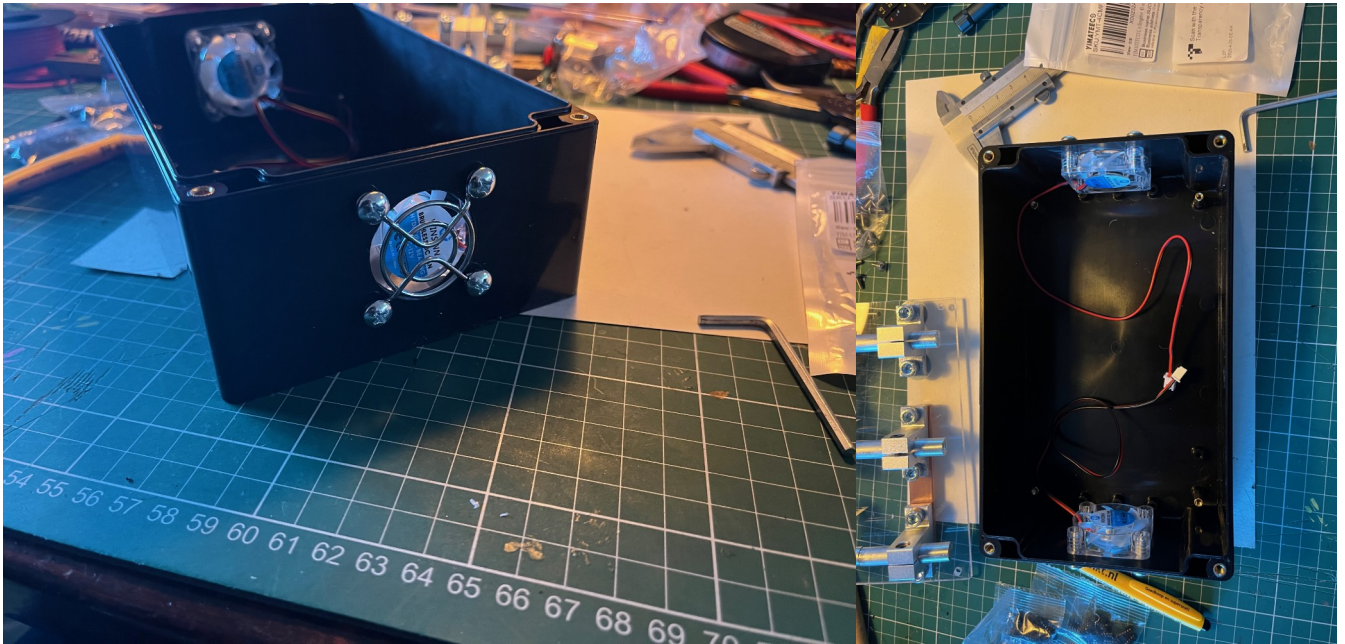
Use the masking tape on the outside of the box to measure and mark the center.



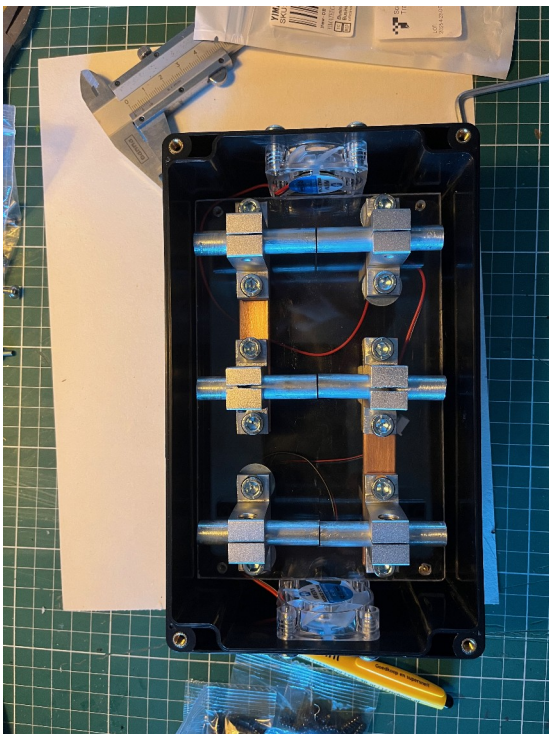
Now you can drill the big hole and then mark where the small holes (4 mm) have to go for the screws to fixate the fans in place.



Now you can screw the fans and fan guards into place (keep in mind the airflow), make sure not to tighten them too strong else they won't be able to turn freely.
Test to see if the fans move freely (keep in mind the airflow)



Now fit the Spark Gap in the box, again make sure the fans can spin freely (you might have to sand the edge down a bit of the plexiglass plate that holds the spark gap)



If the fans can spin freely you can add the cabling to both outer SK10 studs which do not have any copper bar attached to them.

Also holes need to be drilled for the fan cables and the 2 Spark Gap leads you connected previously. No photo's added of that process, you are free to choose which place suits you best for the holes.

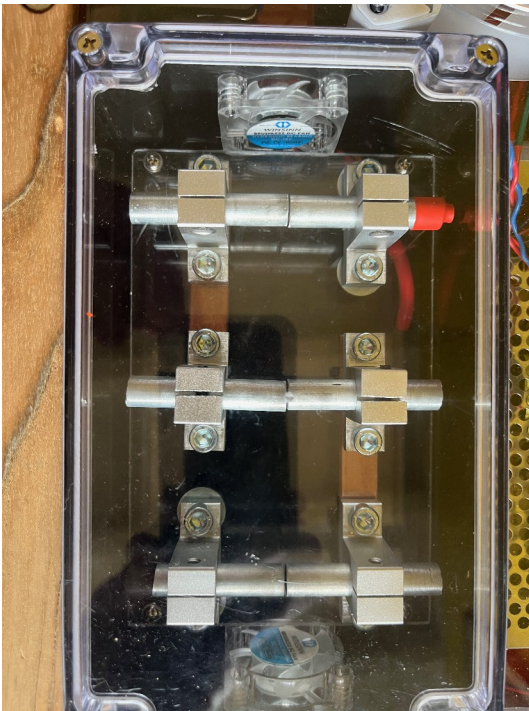
Note: do not place the Spark Gap leads in the same hole as the fan cables. Every cable needs to have it's own individual holes.

Using shrink sleeves to mark the two feeds

When using the black box it is difficult to see which terminal has what color attached to it.

Using a red and black shrink sleeve you can mark the Zinc rods that receive power.

It also helps to quickly identify which poles the short to drain the capacitors when you do Spark Gap maintenance.

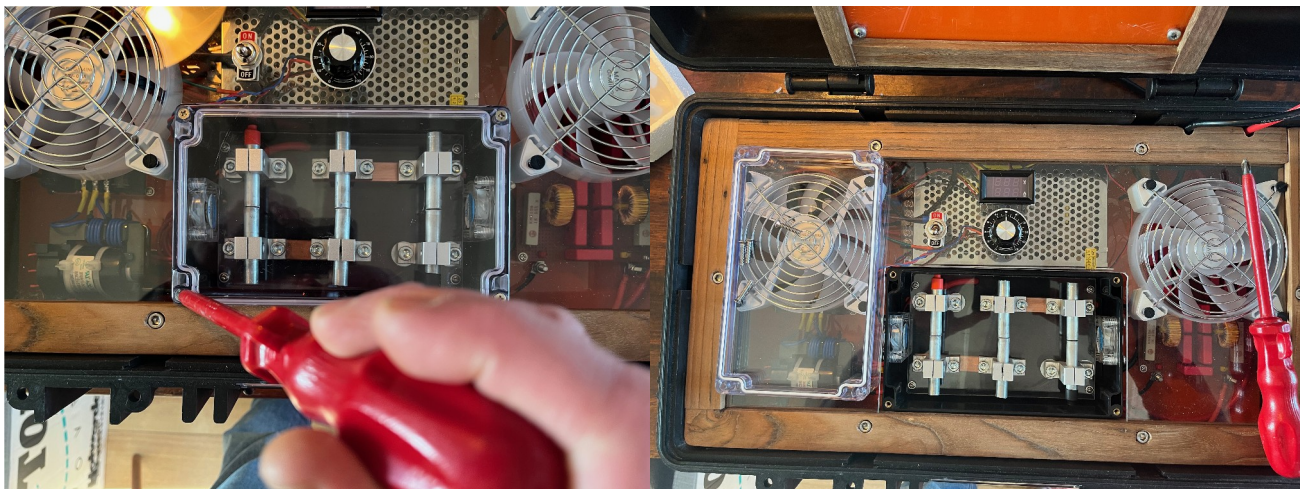


Spark Gap Maintenance

First remove the power cable or power adapter from your machine so you are sure there is no external power source connected.

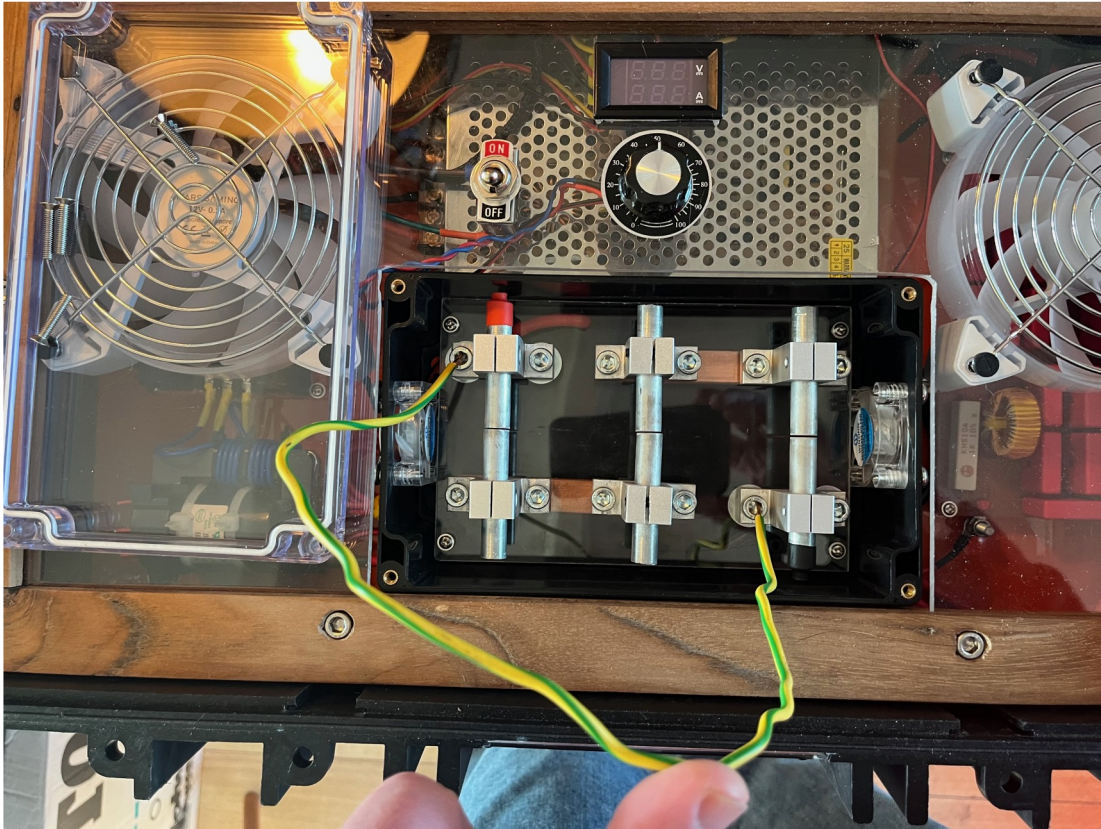


Next open the Spark Gap box.



Before you do any maintenance you have to discharge the capacitor banks.
To do this use a normal piece of electrical wiring to bridge the red and black feeds.

If the machine has been running prior to starting the maintenance this might draw quite a spark, if your lamps are connected they might lighten up during the discharge.



Now that the capacitor banks are drained it is safe to touch the leads and other components in the Spark Gap.

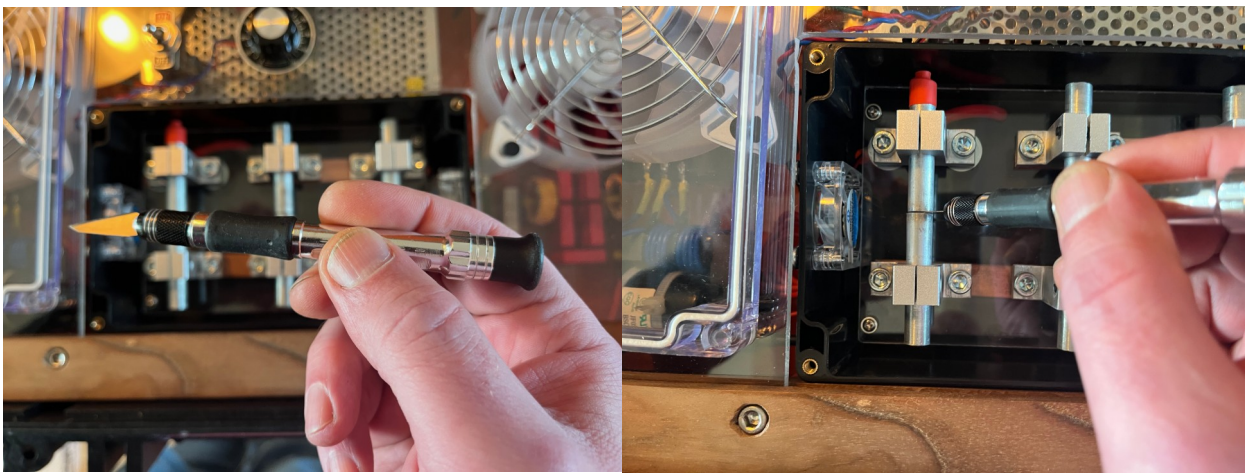
Cleaning the Spark Gap

Discharge the capacitors before doing this step!!! See the first part of Spark Gap Maintenance.

Due to the high voltage going through the Zinc rods at the point where they spark Zinc is slowly corroded away.

Sometimes a ball of Zinc particles might form between the two electrodes of the Spark Gap, normally they burn off quickly but sometimes a ball gets stuck and bridges the two electrodes.

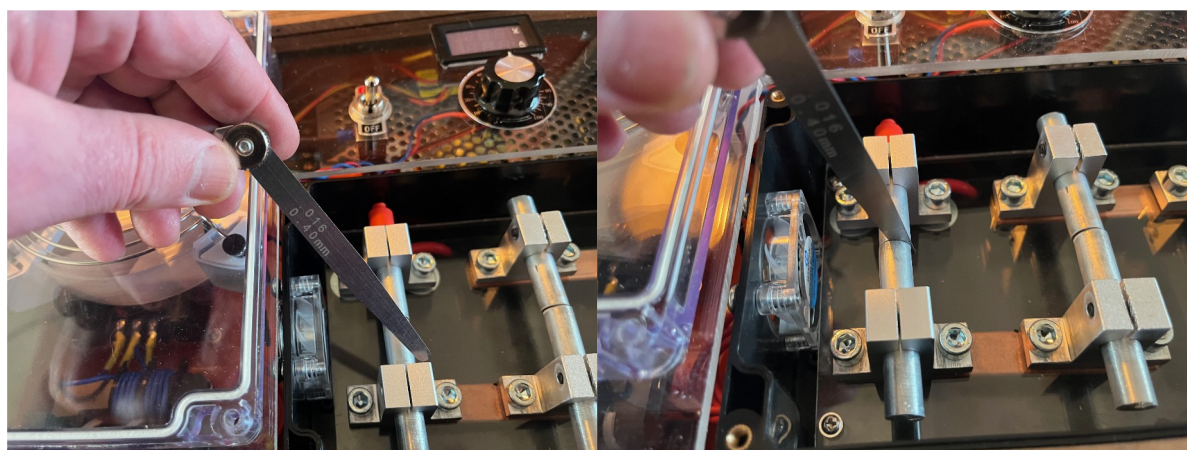
When this happens you'll need to remove it, you can use a boxcutter or X-acto knife to get it out.



Adjusting the Spark Gap distance

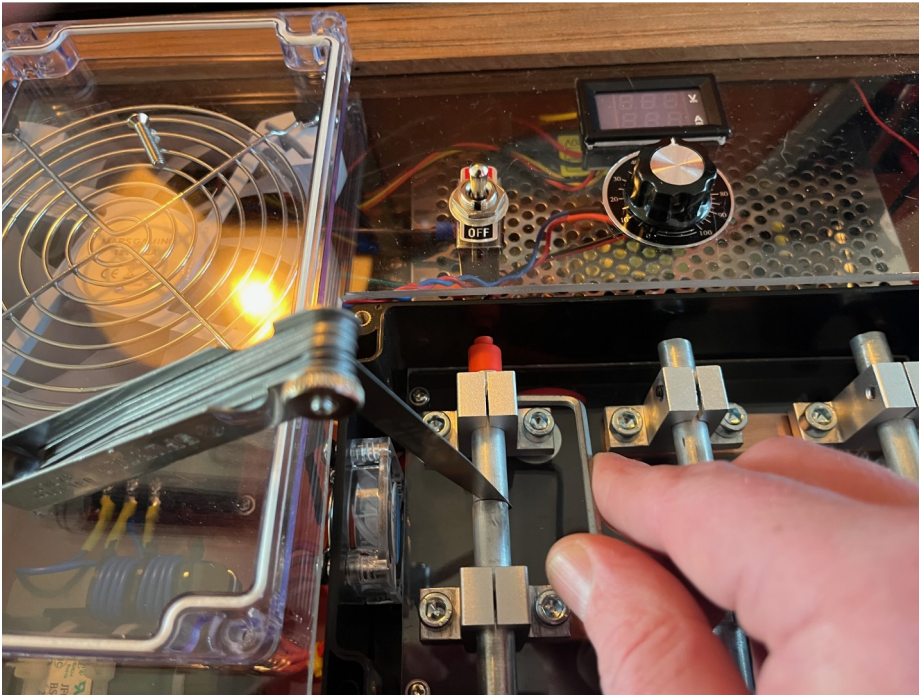
As mentioned before the Zinc electrodes are slowly burned up during machine use, which means that from time to time the spark gap distance needs to be checked to see if these are still between 0.4 and 0.5 mm.

To check this you can measure the distance using a feeler gauge or the thickness of a boxcutter blade.



Wiggle and move the feeler gauge to see how much play there is.

When the feeler gauge moves freely we have to adjust the spark gap and move the electrodes closer to each other.



With an Allen Key you can loosen up one side and bring the rod closer so it fits snugly against the feeler gauge. When you have the desired spark gap distance fasten the clamp.

Close everything up again before connecting the machine back to its power source and starting it up again.